

Week 1 Module – Programming Practice (3rd Year)

Date: 10/06/2026

Language Used

Students can use **either C++ or Java Programming Language** to complete all tasks.

Topics to be Covered:

1. Star Patterns

1. Square 2. Hollow Square 3. Triangle 4. Inverted Triangle

*****	*****	*	*****
*****	* *	**	****
*****	* *	***	***
*****	* *	****	**
*****	*****	*****	*

5. Pyramid 6. Diamond 7. Half Diamond 8. Half Diamond Inverted

*	*	*	*****
***	***	**	****
*****	*****	***	***
*****	*****	****	**
*****	*****	*****	*
	*****	*****	
	*****	***	
	***	**	
	*	*	

2. Number Patterns

1. Square 2. Incrementing 3. Right Triangle 4. Inverted Diamond

1 1 1 1 1	1 2 3 4 5	1	3 3 3 3 3
1 1 1 1 1	1 2 3 4 5	1 2	4 4 4 4
1 1 1 1 1	1 2 3 4 5	1 2 3	5 5 5
1 1 1 1 1	1 2 3 4 5	1 2 3 4	6 6
1 1 1 1 1	1 2 3 4 5	1 2 3 4 5	7

5. Sandwich Pattern

```
3 3 3 3 3
 4 4 4
   5
  4 4 4
3 3 3 3 3
```

6. Incrementing Diamond Pattern

```
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
```

Task Requirements

Students are required to:

- Write the **logic and code** for all given patterns.
- Use **either C++ or Java** for implementation.
- Execute every program and verify the output.

PDF Submission Requirements

Students must prepare a **single PDF document** containing:

1. Handwritten Work

- Handwritten logic/approach for each pattern
- Handwritten code in notebook

2. Executable Code Proof

- Screenshot of successfully executed code/output for every program

GitHub Submission

- Push the final PDF to GitHub.
- **Submission Date: 14/06/2026**
- Repository name format: **3rd Yr Week 1 Module**

Important Instructions

- ✓ Handwritten notes and code are mandatory
- ✓ Output screenshots must be clearly visible
- ✓ Maintain proper sequence and neatness in the PDF
- ✓ Incomplete submissions will not be accepted

1. Square pattern

- Logic:-
1. Start the program
 2. Use an outer loop to print rows
 3. Use an inner loop to print 5 '*' in each row.
 4. After printing one row, move to the next line.
 5. Stop the program.

```
Code:- #include <iostream>
using namespace std;
int main()
{
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= 5; j++) {
            cout << "*" ;
        }
        cout << endl;
    }
    return 0;
}
```

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     for(int i=1;i<=5;i++){
6         for(int j=1;j<=5;j++){
7             cout<<"*";
8         }
9         cout<<endl;
10    }
11
12    return 0;
13 }
```

```
*****
*****
*****
*****
*****
```

=== Code Execution Successful ===

2.] Hollow square pattern

logic:- 1.] start the program

2.] set the size of square as 5.

3.] use nested loops for rows and columns.

4.] print '*' if row is first or last or column is first or last.

5.] otherwise print space.

6.] Move to next line after each row.

7.] stop the program.

Code:-

```
#include <iostream>
```

```
using namespace std;
```

```
int main ()
```

```
{  
    int n=5;
```

```
    for (int i=1; i<=n; i++)
```

```
    {  
        for (int j=1; j<=n; j++)
```

```
        {  
            if (i==1 || i==n || j==1 || j==n)
```

```
                cout << " * ";
```

```
            else
```

```
                cout << "  ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     int n=5;
6     for(int i=1;i<=n;i++){
7         for(int j=1;j<=n;j++){
8             if(i==1||i==n||j==1||j==n)
9                 cout<<"*";
10            else
11                cout<<" ";
12        }
13        cout<<endl;
14    }
15
16    return 0;
17 }
```

```
*****
*  *
*  *
*  *
*****
```

=== Code Execution Successful ===

Snipping

Screenshot
Automatica

3

3.1 Triangle pattern

logic:- (1) Start the program.

2) Use outer loop for rows from 1 to

3) Use inner loop from 1 to current row no.

4) print 'x' in each iteration.

5) move to the next line after each

Q.1) Stop the program.

```
code  
#include <iostream>  
using namespace std;  
int main ()
```

```
{  
    for (int i = 1; i <= 5; i++)
```

```
    {  
        for (int j = 1; j <= i; j++)
```

```
        {  
            cout << " * ";
```

```
        }
```

```
        cout << endl;
```

```
    }  
    return 0;
```

```
}
```



```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     int n=5;
6     for(int i=1;i<=n;i++){
7         for(int j=1;j<=i;j++){
8
9             cout<<"*";
10
11         }
12         cout<<endl;
13     }
14
15     return 0;
16 }
```

```
*
**
***
****
*****
```

=== Code Execution Successful ===

4. Inverted triangle

Logic: ① Start the program.

② print 5 stars in first row.

③ Decrease one star in each row.

④ Continue until only one star remain.

⑤ Stop the program.

Code:

```
1) #include <iostream>
using namespace std;
int main()
```

```
{
    for (int i = 5; i >= 1; i--)
```

```
{
    for (int j = 1; j <= i; j++)
```

```
{
    cout << "*";
```

```
}
```

```
cout << endl;
```

```
}
```

```
return 0;
```

```
}
```

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     int n=5;
6     for(int i=5;i>=1;i--){
7         for(int j=1;j<=i;j++){
8             cout<<"*";
9         }
10        cout<<endl;
11    }
12    return 0;
13 }
```

```
*****
****
***
**
*
```

=== Code Execution Successful ===

(5) Pyramid Pattern.

- 1) Start the program.
- 2) Print required spaces before stars.
- 3) Number of spaces increases row by row.

4.] Print odd number of stars
(4, 3, 5, 7, 9)

5.] Repeat until pyramid is
formed.

6.] Stop the program.

```
#include <iostream>
using namespace std;
int main()
```

```
{ int n = 5;
```

```
  for (int i = 1; i <= n; i++)
```

```
  { for (int j = 1; j <= n - i; j++)
```

```
    cout << " ";
```

```
    for (int k = 1; k <= 2 * i - 1; k++)
```

```
      cout << "*";
```

```
    cout << endl;
```

```
  }
  return 0;
```

```
}
```


// Online C++ compiler to run C++ program online

```
#include <iostream>
using namespace std;
```

```
int main() {
```

```
    int n=5;
```

```
    for(int i=1;i<=n;i++){
```

```
        for(int j=1;j<=n-i;j++){
```

```
            cout<<" ";
```

```
            for(int k=1;k<=2*i-1;k++){
```

```
                cout<<"*";
```

```
            cout<<endl;
```

```
        }
```

```
    return 0;
```

```
}
```

```
^
***
*****
*****
*****
```

=== Code Execution Successful ===

60] Diamond Pattern

- Logic:-
- ① Start the program
 - ② Create upper half using pyramid logic.
 - ③ Create lower half using inverted pyramid logic.
 - 4.] Combine both halves.
 - 5.] Stop the program.

code:-

```
#include <iostream>
using namespace std;
int main()
```

```
{ int n = 5;
```

```
  for (int i = 1; i <= n; i++)
```

```
  { for (int j = 1; j <= n-i; j++)
```

```

1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5     int n=5;
6     for(int i=1;i<=n;i++){
7         for(int j=1;j<=n-i;j++)
8             cout<<" ";
9
10        for(int k=1;k<=2*i-1;k++)
11            cout<<"*";
12
13        cout<<endl;
14    }
15    for(int i=n-1;i>=1;i--){
16        for(int j=1;j<=n-i;j++)
17            cout<<" ";
18
19        for(int k=1;k<=2*i-1;k++)
20            cout<<"*";
21
22        cout<<endl;
23    }
24
25    return 0;
26 }

```

```

*
***
*****
*****
*****
*****
*****
***
*

```

=== Code Execution Successful ===

```

    cout << " ";
    for (int k = 1; k <= 2 * i - 1; k++)
        cout << "* ";
    cout << endl;
}
for (int i = n - 1; i >= 1; i--)
{
    for (int j = 1; j <= n - i; j++)
        cout << " ";
    for (int k = 1; k <= 2 * i - 1; k++)
        cout << "* ";
    cout << endl;
}
return 0;
}

```


7. Half Diamond pattern

- Logic:-
- 1) Print increasing triangle
 - 2) Print decreasing triangle
 - 3) Combine both pattern.
 - 4) Stop the program.

Code:-

```
#include <iostream>
using namespace std;
int main()
{
    for(int i = 1; i <= 5; i++)
    {
        for(int j = 1; j <= i; j++)
            cout << "x";
        cout << endl;
    }
    for(int i = 4; i >= 1; i--)
    {
        for(int j = 1; j <= i; j++)
            cout << "x";
        cout << endl;
    }
    return 0;
}
```

```
#include <iostream>
using namespace std;
int main()
{
    for (int i = 1; i <= 5; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << "*";
        }
        cout << endl;
    }
    for (int i = 4; i >= 1; i--)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << "*";
        }
        cout << endl;
    }

    return 0;
}
```

```
*
**
***
****
*****
****
***
**
*
```

=== Code Execution Successful ===

⑧ Half diamond Inverted

- ① print inverted triangle
- ② print normal triangle
- ③ combine both pattern
- ④ stop the program.

Code:-

```
#include <iostream>
using namespace std;
int main()
```

```
{
    for (int i = 5; i >= 1; i--)
```

```
    {
        for (int j = 1; j <= i; j++)
```

```
            cout << "X" ;
```

```
        cout << endl;
```

```
    }
```

```
    for (int i = 2; i <= 5; i++)
```

```
    {
        for (int j = 1; j <= i; j++)
```

```
            cout << "X" ;
```

```
        cout << endl;
```

```
    }
    return 0;
```

```
}
```

```
// Online C++ compiler to run C++ program online
#include <iostream>
using namespace std;
int main() {
    int n=5;
    for(int i=5;i>=1;i--){
        for(int j=1;j<=i;j++){
            cout<<"*";
            cout<<endl;
        }
        for(int i=2;i<=5;i++){
            for(int j=1;j<=i;j++){
                cout<<"*";
            }
            cout<<endl;
        }

        return 0;
    }
}
```

```
*****
****
***
**
*
**
***
****
*****
```

=== Code Execution Successful ===

}

⑨ Number Square Pattern

- ① start the program
- ② use nested loops for rows and columns
- ③ print number 1 in every position.
- ④ move to next row.
- ⑤ stop the program.

code:-

```
#include <iostream>
using namespace std;
int main()
```

```
{ for(int i = 1; i <= 5; i++)
```

me

```
{  
for(int j=1; j<=5; j++)
```

```
    cout << "1" ;
```

```
    cout << endl;
```

```
}  
return 0;
```

```
}
```

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5
6
7     for(int i=1;i<=5;i++){
8         for(int j=1;j<=5;j++){
9             cout<<"1";
10
11             cout<<endl;
12         }
13
14
15     return 0;
16 }
```

```
11111
11111
11111
11111
11111
```

=== Code Execution Successful ===

10.1 Incrementing Number Pattern

- (1.) Use nested loops
- (2.) Print column numbers in each row.
- (3.) Numbers increase from 1 to 5.
- (4.) Repeat for every row.
- (5.) Stop the program

Code:

```
#include <iostream>
using namespace std;
int main ( )
```

```
{ for (int i = 1; i <= 5; i++)
```

```
{ for (int j = 1; j <= 5; j++)
```

```
    cout << j << "    ";
```

```
    cout << endl;
```

```
}
```

```
return 0;
```

```
}
```



```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5
6
7     for(int i=1;i<=5;i++){
8         for(int j=1;j<=5;j++)
9             cout<<j<<" ";
10
11         cout<<endl;
12     }
13
14
15     return 0;
16 }
```

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

=== Code Execution Successful ===

11] Right triangle Number

logic:-

- 1.] use outer loop for rows
- 2.] use inner loop from 1 to row number
- 3.] print numbers sequentially
- 4.] move to next line
- 5.] stop the program

code:-

```
#include <iostream>
using namespace std;
```

```
int main ( )
```

```
{
```

```
    for (int i = 1; i <= 5; i++)
```

```
    {
```

```
        for (int j = 1; j <= i; j++)
```

```
            cout << j << "    ";
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4 int main() {
5
6
7     for(int i=1;i<=5;i++){
8         for(int j=1;j<=i;j++)
9             cout<<j<<" ";
10
11         cout<<endl;
12     }
13
14
15     return 0;
16 }
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

=== Code Execution Successful ===

130] Inverted diamond

Logic:- Start a value variable at 3.

2.) Row decrease from 5 to 1

3.) Print current number according to row length

4.) Increase number after every row.

Code:-

```
#include <iostream>
using namespace std;
int main()
```

```
{
    int num = 3;
```

```
    for(int i = 5; i >= 1; i--)
```

```
    {
```

```
        for(int j = 1; j <= i; j++)
```

```
        {
```

```
            cout << num << " ";
```

```
        }
```

```
        num++;
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

```
1 // Online C++ compiler to run C++ program online
2 #include <iostream>
3 using namespace std;
4
5 int main() {
6     int num = 3;
7     for (int i = 5; i >= 1; i--) {
8         for (int j = 1; j <= i; j++) {
9             cout << num;
10        }
11        cout << endl;
12        num++;
13    }
14    return 0;
15 }
```

33333
4444
555
66
7

=== Code Execution Successful ===

14] Sandwich pattern -
Logic!-

Start with number 3 and
print 5 times

② ↑ the number and
decrease the count by 2 in
each row

• 3 → 5 times
• 4 → 3 times
• 5 → 1 times

3] Add leading spaces to center the pattern.

4.] After reaching the middle(s), print pattern in reverse.

• 4 → 3 times

• 3 → 5 times

Code:-

```
#include <iostream>
using namespace std;
int main()
```

```
{
    int space = 0;
```

```
    for (int i = 3; i <= 5; i++)
```

```
    {
        for (int s = 0; s < space; s++)
```

```
            cout << "  ";
```

```
        for (int j = 5; j >= 2 * (i - 3) + 1; j--)
```

```
        {
            cout << i << "  ";
```

```
            cout << endl;
```

```
            space++;
```

```
        }
        space = 1;
```

```
        for (int i = 4; i <= 3; i--)
```

```
        {
            for (int s = 0; s < space; s++)
```

```
                cout << "  ";
```

```
            for (int j = 5; j >= 2 * (i - 3) + 1; j--)
```



```

#include <iostream>
using namespace std;
int main()
{
    int spaces = 0;
    for(int i = 3; i <= 5; i++)
    {
        for(int s = 0; s < spaces; s++)
            cout << " ";
        for(int j = 5; j >= 2*(i-3)+1; j--)
            cout << i << " ";
        cout << endl;
        spaces++;
    }
    spaces = 1;
    for(int i = 4; i >= 3; i--)
    {
        for(int s = 0; s < spaces; s++)
            cout << " ";
        for(int j = 5; j >= 2*(i-3)+1; j--)
            cout << i << " ";
        cout << endl;
        spaces--;
    }
    return 0;
}

```

```

3 3 3 3 3
4 4 4
5
4 4 4
3 3 3 3 3

```

=== Code Execution Successful ===

```

    cout << i << "
    cout endl;
    spaces --;
}
return 0;
}

```

15) Incrementing Diamond pattern

Logic:-

- ① print upper half from 3 to 7.
- ② Number of elements increases from 1 to 5.
- ③ print lower half from 6 to 3
- ④ Number of element decreases from 4 to 1

Code:-

```

#include <iostream>
using namespace std;
int main()
{
    int num = 3;
    for (int i = 1; i <= 5; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << num << " ";
        }
        cout << endl;
        num++;
    }
    num = 6;
    for (int i = 4; i >= 1; i--)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << num << " ";
        }
        cout << endl;
        num--;
    }
    return 0;
}

```

```

2 #include <iostream>
3 using namespace std;
4 int main()
5 {
6     int num=3;
7     for(int i=1;i<=5;i++)
8     {
9         for(int j=1;j<=i;j++)
10        {
11            cout<<num<<" ";
12        }
13        cout<<endl;
14        num++;
15    }
16    num=6;
17    for(int i=4;i>=1;i--)
18    {
19        for(int j=1;j<=i;j++)
20        {
21            cout<<num<<" ";
22        }
23        cout<<endl;
24        num--;
25    }
26    return 0;
27 }

```

```

3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3

```

=== Code Execution Successful ===